



Dynamic depth quantum approximate optimization algorithm for solving constrained shortest path problem

Rakesh Saini¹ · Nora Mohamed¹ · Saif Al-Kuwari¹ · Ahmed Farouk^{1,2}

Received: 18 November 2025 / Accepted: 19 August 2026
© The Author(s) 2026

Abstract

The Quantum Approximate Optimization Algorithm (QAOA) has emerged as a promising approach for solving NP-hard combinatorial optimization problems on noisy intermediate-scale quantum (NISQ) hardware. However, its performance critically depends on the choice of circuit depth—a parameter that must be specified a priori without clear guidance. In this paper, we introduce a variant of QAOA, called the Dynamic Depth Quantum Approximate Optimization Algorithm (DDQAOA), that addresses the challenge of pre-selecting a fixed circuit depth. Our method adaptively expands circuit depth, starting from $p = 1$ and progressing up to $p = 10$, by transferring learned parameters to deeper circuits based on convergence criteria. We tested this approach on 100 instances of the Constrained Shortest Path Problem (CSPP) at 10-qubit and 16-qubit scales, and on 20 additional instances at the 22-qubit scale. DDQAOA achieved competitive approximation ratios and success probabilities with substantially fewer CNOT evaluations than standard QAOA at $p = 10$ and $p = 15$. In particular, while standard QAOA at $p = 15$ achieved results close to our approach, it used 217%, 159.3%, and 315% more CNOT gates for 10-qubit, 16-qubit, and 22-qubit instances, respectively. These results demonstrate that convergence-driven DDQAOA, building on the INTERP parameter-transfer protocol (Zhou et al. 2020), offers a practical route to applying QAOA to constrained NP-hard problems on near-term devices.

Keywords Quantum approximate optimization algorithm · Constrained shortest path problem · Ising formulation · Quadratic unconstrained binary optimization

1 Introduction

Combinatorial optimization problems arise in a wide range of applications, from large-scale integrated circuit design and drug discovery to financial portfolio management. These problems are not only of great theoretical interest but also central to numerous applications in real-world decision-making systems, making them fundamental to both scientific research and industrial practice (Goto et al. 2019;

Tong 2021; Ramos-Figueroa et al. 2020). However, solving these problems on classical computers is difficult due to the exponentially growing solution space and the ubiquity of local optima (Wolsey and Nemhauser 1999; Arora and Barak 2009).

Quantum computing has emerged as a promising approach to achieve exponential computational acceleration by exploiting quantum phenomena (Arute et al. 2019; Daley et al. 2022; Zhong et al. 2020). However, current quantum hardware faces significant limitations in qubit connectivity, error susceptibility, and qubit counts, placing it firmly within the NISQ era (Bharti et al. 2022). Despite these limitations, NISQ devices provide a platform for developing and testing early generations of quantum algorithms that can potentially demonstrate quantum advantages (Preskill 2018; Cerezo et al. 2021).

Among these algorithms is QAOA (Farhi et al. 2014), which has become one of the most popular approaches for solving combinatorial optimization problems with its

✉ Rakesh Saini
rasa68842@hbku.edu.qa

¹ Qatar Center for Quantum Computing, College of Science and Engineering, Hamad Bin Khalifa University, Doha, Qatar

² Department of Computer Science, Faculty of Computers and Artificial Intelligence, Hurgada University, Hurgada, Egypt

hybrid quantum-classical approach making it particularly well-suited for NISQ hardware (Farhi et al. 2014; Blekos et al. 2024; Harrigan et al. 2021). However, as problem complexity increases, achieving enhanced solution quality often requires deeper quantum circuits, making current experimental implementations impractical due to limitations in decoherence times and gate error rates (Wurtz and Love 2021; Zhou et al. 2020; Zhu et al. 2023; Ebadi et al. 2022). While QAOA performance is restricted at lower depths, the higher depths needed to reach optimal solutions exceed the capabilities of NISQ devices and experimental techniques (Zou 2025; Montañez-Barrera and Michielsen 2025).

A fundamental challenge in QAOA implementation is selecting an appropriate circuit depth. While deeper circuits generally yield higher solution quality, they also incur substantially greater classical-optimization and two-qubit-gate overhead, so the practical challenge is to balance solution quality against this overhead rather than to locate a single “optimal” depth. In particular, fixed-depth QAOA implementations suffer from fundamental underparametrization as problem density increases since the constraint-to-variable ratio induces performance degradation that cannot be overcome without increasing circuit depth. Although fixed circuit depths may exhibit insensitivity to problem size for certain problems, such as Max-Cut, the general requirement that the depth p needed to reach a given solution quality grows with problem size means that fixed-depth approaches either underperform by using insufficient depth or become computationally wasteful by using excessive depth for simpler instances (Crooks 2018; Akshay et al. 2022; Sureshbabu et al. 2024; Truger et al. 2024).

To address these limitations, several adaptive approaches have been proposed (Tyagin et al. 2025; Cheng et al. 2024; Yanakiev et al. 2024; Yu et al. 2023). The most prominent among these is ADAPT-QAOA (Zhu et al. 2022), which provides a systematic method for iteratively building problem-tailored ansätze by selecting operators from a predefined pool based on gradient criteria, thereby improving performance while reducing both the parameter count and circuit complexity. However, this method incurs computational overhead from repeated gradient evaluations at each iteration, with the cost scaling with the size of the operator pool. More broadly, adaptive ansatz construction has been extensively studied in the variational quantum eigensolver (VQE) context, sharing the conceptual goal of circuit-depth efficiency. ADAPT-VQE (Grimsley et al. 2019) iteratively grows the ansatz by selecting operators from a predefined pool based on gradient magnitudes, and its

qubit-efficient variant (Tang et al. 2021) reduces the operator pool to single-qubit excitations. These methods adaptively construct the circuit structure by choosing which operators to append; our DDQAOA, in contrast, keeps the standard QAOA ansatz fixed and adaptively determines only the number of layers, using a convergence-driven depth-increment mechanism combined with parameter interpolation. In this sense, DDQAOA is complementary to operator-selection approaches and closely related to interpolation-based parameter-transfer methods (see Section 3). Complementing these advances, parameter transfer methods (Montañez-Barrera and Michielsen 2025; Sureshbabu et al. 2024) enable warm-starting QAOA by transferring optimized parameters from smaller or related problem instances to target problems, a process that depends primarily on structural properties such as graph degree and initialization states. Moreover, a layerwise variant of QAOA has been developed (Lee et al. 2024), which trains the layer parameters sequentially; however, this approach can get stuck in suboptimal solutions due to its restricted search space. Warm-started QAOA (Egger et al. 2021) that initializes the QAOA in a quantum state based on a classical approximate solution, which means that the performance depends on the solution provided by classical pre-computation. These approaches address specific aspects of QAOA but do not resolve the challenge of selecting an appropriate circuit depth a priori, which is a critical bottleneck.

1.1 Contributions

In this paper, we present DDQAOA, a variant of QAOA that resolves the challenge of pre-selecting a fixed circuit depth, thereby improving both the performance and practical applicability of the algorithm. The key contributions of this work are as follows:

- Introducing DDQAOA, A convergence-driven variant of interpolation-based QAOA, in which depth is increased upon detection of local convergence rather than via a predetermined schedule.
- A parameter-transfer rule building on the interpolation-based protocol of Zhou et al. (2020), with transfer coefficients for the $p = 1 \rightarrow 2$ transition motivated by adiabatic scheduling.
- A comprehensive demonstration of the method’s effectiveness on CSPP, which is an NP-hard problem. A non-trivial benchmark for constrained optimization.
- Competitive approximation ratios at a fraction of the cumulative CNOT budget of standard fixed-depth QAOA ($p = 10, 15$).

1.2 Organization

The remainder of this paper is organized as follows. Section 2 provides a review of the foundational QAOA framework and the mathematical formulation of the Constrained Shortest Path Problem. Section 3 presents a detailed description of our DDQAOA. In Section 4, we describe the experimental design, including the problem instances, baseline methods, and performance metrics used for our evaluation. In section 5, we present and analyze the results of our simulations, demonstrating the advantages of DDQAOA over the standard QAOA. In Section 6, we summarize our contributions and outline directions for future research.

2 Preliminaries

In this section, we briefly review the QAOA framework, including its mathematical formulation and the role of cost and mixer Hamiltonians in the optimization process. We then present the specific formulation of the constrained shortest path problem and discuss its encoding as a Quadratic Unconstrained Binary Optimization (QUBO) problem suitable for quantum algorithms.

2.1 Quantum approximate optimization algorithm

Many interesting real-world problems can be framed as combinatorial optimization problems (Papadimitriou and Steiglitz 1998; Korte and Vygen 2008). Consider an optimization problem defined on N -bit binary strings $z = z_1 z_2 \dots z_N \in \{+1, -1\}^N$, where the goal is to determine a string that maximize/minimize a given objective function $C(z) = \sum_j C_j(z)$. The mapping of $C(z)$ to the Ising Hamiltonian H_C will be easy if $C_j(z)$ depends on not more than two of the z_i , whereas if it depends on three or more z_i , $C(z)$ may still be mapped onto the Ising Hamiltonian, potentially at the expense of introducing additional auxiliary variables (Chancellor et al. 2017). The Ising Hamiltonian is diagonal in the Pauli-Z basis, and the ground state energy, denoted by $E_C^{(0)}$, corresponds (up to an irrelevant constant) to the minimum of $C(z)$.

In QAOA, the quantum circuit is initialized in the equal superposition of all possible bit-strings $|+\rangle^{\otimes N}$, which is achieved by applying the Hadamard gates $H^{\otimes N}$ to $|0\rangle^{\otimes N}$. Next, a variational ansatz created using the problem Hamiltonian H_C and a mixer Hamiltonian $H_M = \sum_{j=1}^N \sigma_j^x$

alternately with controlled durations is applied, thus achieving the QAOA state ψ :

$$|\psi_p(\gamma, \beta)\rangle = e^{-i\beta_p H_M} e^{-i\gamma_p H_C} \dots e^{-i\beta_1 H_M} e^{-i\gamma_1 H_C} |+\rangle^{\otimes N}, \quad (1)$$

which is parameterized by $2p$ variational parameters γ_i and β_i where $i = 1, 2, \dots, p$ is to be optimized by a classical algorithm. The goal of QAOA is to find such optimal parameters γ^* and β^* , so that the expected value

$$F_p(\vec{\gamma}, \vec{\beta}) = \langle \psi_p(\vec{\gamma}, \vec{\beta}) | H_C | \psi_p(\vec{\gamma}, \vec{\beta}) \rangle, \quad (2)$$

is maximized (or minimized). QAOA optimization typically starts with an initial guess for the parameters and then performs simplex- or gradient-based optimization to obtain the optimal parameters for the next step. The expectation value is estimated by repeatedly preparing the state $|\psi_p(\vec{\gamma}^*, \vec{\beta}^*)\rangle$ at the optimized parameters. Once the optimization ends, the approximation ratio at the final optimal parameters is calculated to benchmark the performance of the QAOA:

$$r = \frac{F_p(\vec{\gamma}^*, \vec{\beta}^*)}{C_{\max}}. \quad (3)$$

QAOA starts with the lowest-energy eigenstate of the mixer Hamiltonian and, according to the adiabatic theorem, the system should evolve toward the lowest-energy eigenstate of the cost Hamiltonian. QAOA is a digitized/Trotterized approximation of adiabatic quantum computation that discretizes the continuous adiabatic evolution into alternating unitary operations. In our work, we seek the ground state of the cost Hamiltonian; hence, we aim to minimize its expected value.

2.2 Constrained shortest path problem

The CSPP is a fundamental NP-hard problem, even when considering only a single resource constraint, which involves finding a minimum cost path between two points while satisfying a number of resource constraints (Garey and Johnson 2002; Handler and Zang 1980). This problem serves as an excellent testbed for QAOA benchmarking. The problem, visualized in Fig. 1, involves finding a minimum-cost path between a source node s and target node t in a bi-directional graph, subject to resource constraints.

Given a bi-directed graph $G = (V, E)$ with vertex set $V = \{1, 2, \dots, n\}$ representing possible locations, and edge set $E = \{(i, j) \mid i, j \in V, i \neq j\}$ representing the transition between any two locations. Each edge $((i, j) \in E)$ in the graph is associated with:

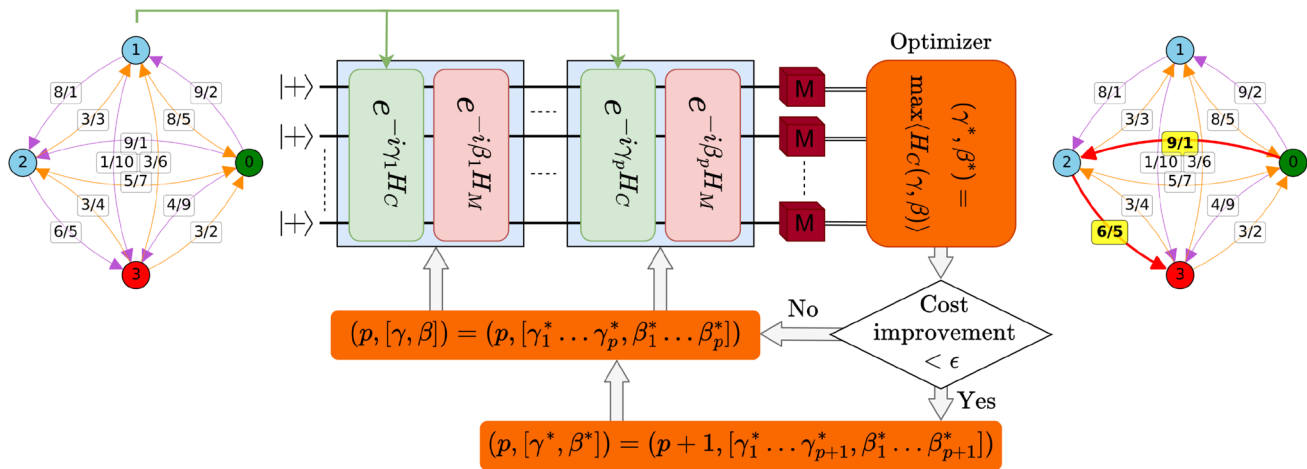


Fig. 1 The DDQAOA quantum circuit schematic applied to 4-node CSPP instance. Each box in the circuit contains the cost Hamiltonian operator $e^{-i\gamma H_C}$ and mixer operator $e^{-i\beta H_M}$, H_C corresponds to the problem. The algorithm iteratively optimizes (γ, β) at depth p using a classical optimizer to maximize $\langle H_C(\gamma, \beta) \rangle$. Here, the DDQAOA starts with

- A non-negative travel cost value $c_{ij} \in \mathbb{R}^+$
- A non-negative resource consumption value $r_{ij} \in (\mathbb{R}^+)^M$ with M being a positive integer denoting the number of different resources (e.g., fuel, time, distance, etc.). In this paper, $M = 1$, which means that the problem we consider here is equivalent to the Weight Constrained Shortest Path Problem (Ren et al. 2023).

Let v_s and v_t denote the initial vertex and target vertex, respectively. Let $P(v_1, v_\ell) := \{v_1, v_2, \dots, v_\ell\}$ denote a path that consists of a list of vertices with each pair of adjacent vertices v_k, v_{k+1} , $k \in \{1, 2, \dots, \ell - 1\}$ connected by an edge $(v_k, v_{k+1}) \in E$. We refer to $P(v_1, v_\ell)$ simply as P when no confusion arises. For a path $P(v_1, v_\ell)$, let

$c(P) := \sum_{k=0}^{\ell-1} c(v_k, v_{k+1})$ represent the path cost, which is the cumulative cost of the edges that are present in the path

P . Similarly, let $r(P) := \sum_{k=0}^{\ell-1} \vec{r}(v_k, v_{k+1})$ denote the path

resource cost, which describes the total amount of resource consumed when moving from v_1 to v_ℓ along P . Finally, let $r_{\text{limit}} \in \mathbb{R}^+$ denote the resource limit.

The objective is to find a path P from the source v_s to the target v_t that minimizes the total cost:

$$\min \sum_{(i,j) \in P} c_{ij} \quad (4)$$

$p = 1$. When the cost improvement between consecutive iterations falls below threshold ϵ , the depth p is increased by 1 and the optimized p parameters are interpolated to $p + 1$ parameters $(\gamma_1^* \dots \gamma_{p+1}^*, \beta_1^* \dots \beta_{p+1}^*)$. The solution shown in the right displays the optimal CSPP solution from node 0 to node 3, highlighted with a bold red line

while satisfying the resource constraint:

$$\sum_{(i,j) \in P} r_{ij} \leq r_{\text{limit}} \quad (5)$$

In addition to the straightforward application of CSPP (i.e., shortest path subject to a resource limit), the CSPP and its variants naturally arise as auxiliary problems in column generation schemes for air-cargo route planning (Derigs et al. 2009), flight-planning optimization (Graves et al. 1993), crew-pairing (Lavoie et al. 1988; Muter et al. 2013), the tail-assignment problem in aircraft scheduling (Grönkvist 2006), day-to-day crew operations (Stojković et al. 1998), and crew-rostering (Gamache et al. 1999). Some variants of the CSPP include non-additive objective functions (Reinhardt and Pisinger 2011), probability functions defined over the network (Bertsekas and Tsitsiklis 1991), forbidden-path constraints (Villeneuve and Desaulniers 2005), and replenishment arcs that reset resource counters (Smith et al. 2012).

2.2.1 Quadratic Program of CSPP

To represent the CSPP as a binary quadratic program, we use decision variables $x_{ij} \in \{0, 1\}$ that indicate whether edge (v_i, v_j) is included in the optimal path. If it is 1, then there exists an edge from i to j ; otherwise, it is 0. The quadratic program of CSPP can be formulated as:

$$\text{minimize } \sum_{(i,j) \in E} c_{ij} x_{ij} \quad (6)$$

Subject to the following constraints:

$$\sum_{(s,j) \in E} x_{sj} = 1; \sum_{(i,s) \in E} x_{is} = 0 \quad \forall (i,j) \in V \quad (7)$$

$$\sum_{(i,t) \in E} x_{it} = 1; \sum_{(t,j) \in E} x_{tj} = 0 \quad \forall (i,j) \in V \quad (8)$$

$$\sum_{(i,j) \in E} x_{ij} = \sum_{(j,i) \in E} x_{ji} \quad \forall i \in V \setminus \{s, t\} \quad (9)$$

$$\sum_{(i,j) \in E} r_{ij} x_{ij} \leq r_{\text{limit}} \quad (10)$$

Equation 7 imposes the source constraints, which state that there should be an edge living source and there should be no edges entering the source. Also, Eq. 8 applies the target constraints: only one edge can enter the target, and no edge leaves from the target. Moreover, flow conservation constraints are imposed by Eq. 9, stating that for the vertices except s and t , the number of edges entering a vertex should be equal to the number of edges leaving the vertex. Finally, Eq. 10 bounds the optimal path P not to use the resources more than r_{limit} .

The flow-conservation constraints (Eq. 9) are necessary but not by themselves sufficient to eliminate isolated closed loops disconnected from the s - t path: a solution that includes such a loop at intermediate vertices trivially satisfies flow balance at those vertices. However, because all edge costs $c_{ij} > 0$ and all resource values $r_{ij} > 0$, any solution containing an isolated loop has strictly higher objective value (Eq. 6) and strictly higher resource consumption (Eq. 10) than the corresponding loop-free solution. Together with the penalty structure of Eqs. 12–15, this guarantees that isolated loops increase the QUBO energy and hence cannot be optimal. We have additionally verified this empirically: across all 100 problem instances used in Sections 4 and 5, none of the optimal or near-optimal solutions recovered by DDQAOA contained isolated sub-loops.

2.2.2 Mapping to a QUBO

CSPP cannot be solved directly by QAOA, which are designed to find the ground state of unconstrained combinatorial optimization problems. Therefore, the problem must be transformed into an unconstrained form, achieved by reformulating it as a QUBO. The QUBO formalism is a cornerstone of both classical and quantum annealing, as well as variational algorithms, and seeks to minimize a function of the form:

$$f(x) = x^T Q x = \sum_{i,j} Q_{ij} x_i x_j \quad (11)$$

where x is a vector of binary variables and Q is a real-valued square matrix encoding the problem's structure.

This transformation involves integrating the constraints from Eqs. 7, 8, 9, 10 into an objective function as quadratic penalty terms. This results in a single QUBO objective function, which is a weighted sum of the original cost and the following penalties:

$$H_{\text{CSPP}} = H_{\text{cost}} + H_{\text{resource}} + H_{\text{flow}} \quad (12)$$

$$H_{\text{cost}} = \sum_{(i,j) \in E} c_{ij} x_{ij} \quad (13)$$

$$H_{\text{resource}} = \rho \left(\sum_{(i,j) \in E} r_{ij} x_{ij} - r_{\text{limit}} \right)^2 \quad (14)$$

$$H_{\text{flow}} = \lambda \left[\left(\sum_{(s,j) \in E} x_{sj} - 1 \right)^2 + \left(\sum_{(i,t) \in E} x_{it} - 1 \right)^2 + \left(\sum_{(j,s) \in E} x_{js} \right)^2 + \left(\sum_{(t,j) \in E} x_{tj} \right)^2 + \sum_{v \in V \setminus \{s,t\}} \left(\sum_{(i,v) \in E} x_{iv} - \sum_{(v,j) \in E} x_{vj} \right)^2 \right] \quad (15)$$

where ρ and $\{\lambda\}$ are penalty coefficients that have a value of zero when its corresponding constraint is satisfied and a very large positive value (must be set large enough to dominate the cost term) when violated.

The final step in this mapping is to convert the classical QUBO expression into a quantum mechanical Ising Hamiltonian. This is done by the standard transformation $x_i = (1 - s_i)/2$, where $s_i \in \{-1, +1\}$ are spin variables corresponding to Pauli-Z operators Z_i acting on the i -th qubit. This yields the final problem Hamiltonian:

This penalty formulation can be systematically converted to standard QUBO form:

$$H_{\text{QUBO}} = \min \quad x^T Q x + g^T x + c_0 \quad (16)$$

where the QUBO matrix Q , linear vector g , and constant c_0 encode the cost function, resource constraints, and flow conservation requirements.

Finally, the QUBO formulation is converted to an Ising Hamiltonian through the standard transformation $x_i = (1 - s_i)/2$ where $s_i \in \{-1, +1\}$ are spin variables corresponding to Pauli-Z operators Z_i . This process results in an Ising problem Hamiltonian of the general form:

$$H_{\text{Ising}} = E_0 \mathbb{I} + \sum_i h_i Z_i + \sum_{i < j} J_{ij} Z_i Z_j \quad (17)$$

where the parameters $\{h_i\}$ (local fields), $\{J_{ij}\}$ (couplings), and E_0 (constant) are derived from the QUBO coefficients. The ground state of this Hamiltonian corresponds to the optimal solution of the CSPP.

3 Dynamic depth quantum approximate optimization algorithm

DDQAOA extends the interpolation-based parameter-transfer strategy of Zhou et al. (INTERP) (Zhou et al. 2020) by replacing the predetermined depth schedule with a convergence-driven rule for depth increments. Our motivation for developing DDQAOA stems from three observations. First, in fixed-depth QAOA, one must decide how many layers to use. For this, the user tries different numbers of layers until they find an optimal or sub-optimal number of layers. If the number of layers is small, the circuit will be underparameterized and will not converge to an optimal solution. Conversely, if a large number of layers is provided, unnecessary overhead will be introduced (Vijendran et al. 2024; Akshay et al. 2022; Pan et al. 2022). Second, if the number of layers is fixed throughout the optimization, even at a well-chosen depth, one has to perform a significant amount of CNOT gate simulation. As CNOT gates induce noise, minimizing the number of cost unitaries in the QAOA circuit could be valuable (Long et al. 2024). Third, the trends of the optimal parameters (γ^*, β^*) should be increasing and decreasing, respectively, with increasing adiabaticity; however, fixed-depth QAOA yields random trends in the learned parameters (Ruan et al. 2025).

In general, increasing the QAOA depth improves solution quality, with convergence to the optimum guaranteed as $p \rightarrow \infty$; however, the depth required grows with problem size and deeper circuits incur higher optimization overhead. Hence, in our DDQAOA, we first initialize the optimization process with a minimum number of QAOA layers, i.e., 1, and then systematically increase the depth once the expectation value converges to a stable point. We used a dual convergence detection mechanism. The first mechanism detects a plateau by tracking the expectation-value optimization: when the energy improvement falls below a convergence tolerance and no significant improvement occurs for consecutive iterations, it indicates the exhaustion of optimization capacity at the current circuit depth. The second mechanism is an enhanced convergence check that analyzes variance across recent iterations to distinguish true convergence from oscillatory behavior near local minima.

Formally, let E_t denote the expectation value at optimization step t , E_{best} the best value observed so far at the current depth, k the patience window, ϵ the energy-improvement threshold, and σ the variance threshold. At each step after reaching the patience window, the algorithm evaluates two conditions:

Condition 1 (plateau detection):

$$E_{\text{best}} - \min_{s \in [t-k+1, t]} E_s < \epsilon \quad (18)$$

Condition 2 (variance-based convergence):

$$\text{Var}(\{E_{t-[k/2]+1}, \dots, E_t\}) < \sigma \quad (19)$$

If either condition holds, the depth is incremented ($p \leftarrow p + 1$) and the parameters at the new depth are obtained by the transfer scheme of Eqs. 20–21 (for $p = 1 \rightarrow 2$) or the interpolation scheme described in Eq. 22 (for $p \geq 2$). Condition 2 is a secondary safeguard that prevents the algorithm from declaring convergence based on small oscillations near a local minimum. In all experiments reported in Sections 4 and 5, we set $\epsilon = 10^{-4}$, $\sigma = 10^{-6}$, and $k = 50$.

The parameter p_{max} is a user-specified upper bound on the circuit depth and is not an algorithmic limitation of DDQAOA. It can be chosen by the practitioner according to the requirements of the problem at hand: for example, based on a desired solution-quality target, an available time or compute budget, the depth at which further refinement yields diminishing returns for a given problem class, or hardware-imposed constraints such as qubit coherence times and cumulative gate-error budgets on a specific NISQ device. The algorithm itself imposes no fixed ceiling; p_{max} can be increased or decreased without modifying the convergence-driven adaptation mechanism described above. In the experiments reported here we set $p_{\text{max}} = 10$, which was sufficient for the CSPP instances considered: the algorithm terminated via the convergence criteria of Eqs. 20–21 at depths $p \leq 10$ in every one of the 100 instances, so the $p_{\text{max}} = 10$ cap was never the active stopping condition.

Next, when transferring the learned optimized parameter from p to $p + 1$, we used an adaptive interpolation method. For $p = 1$ to $p = 2$:

$$\gamma^{(2)} = [\gamma_1^{(1)}, 1.2 \cdot \gamma_1^{(1)}] \quad (20)$$

$$\beta^{(2)} = [\beta_1^{(1)}, 0.8 \cdot \beta_1^{(1)}] \quad (21)$$

The choice of transfer coefficients (1.2 for γ and 0.8 for β) are motivated by the adiabatic-schedule structure of QAOA parameters (Zhou et al. 2020; Ruan et al. 2025). Theoretical and empirical analyses of optimal QAOA parameters suggest that γ increases approximately linearly with layer index while β decreases, consistent with a discretized adiabatic schedule that progressively strengthens the cost Hamiltonian and weakens the mixer. Under an idealized linear-ramp schedule, the $p = 1 \rightarrow p = 2$ transition would give $\gamma_2 \approx 2\gamma_1$ and $\beta_2 \approx 0$; however, such aggressive extrapolation risks overshooting the optimal values for a shallow ansatz and

destabilizing the subsequent optimization. We therefore adopt moderate coefficients (1.2 and 0.8) that approximate the correct monotonic trend while remaining close to the previously optimized parameters, so that the interpolation-then-reoptimize step starts within the basin of attraction of the $p = 2$ local optimum. For $p \geq 2$, let $\gamma^{(p)} = (\gamma_1, \dots, \gamma_p)$ and $\beta^{(p)} = (\beta_1, \dots, \beta_p)$ denote the optimized parameters at depth p . We map these to uniformly spaced grid points $s_k = (k-1)/(p-1)$, $k = 1, \dots, p$, and construct interpolants J_γ and J_β over this grid. The warm-start parameters at depth $p+1$ are then obtained by evaluating the interpolants at the finer grid $s'_j = (j-1)/p$, $j = 1, \dots, p+1$:

$$\gamma_j^{(p+1)} = J_\gamma(s'_j), \quad \beta_j^{(p+1)} = J_\beta(s'_j), \quad j = 1, \dots, p+1. \quad (22)$$

For $p \in \{2, 3\}$ the interpolants are linear (cubic splines are not well-conditioned on fewer than four points); for $p \geq 4$ we use cubic spline interpolation, which provides a smoother schedule and better approximates the continuous adiabatic limit. This interpolation-then-reoptimize step follows the INTERP (Zhou et al. 2020); DDQAOA differs from INTERP only in that the step is triggered by the convergence criterion of Eqs. 18–19 rather than at every iteration of an outer loop. Algorithm 1 provides a pseudocode for our DDQAOA and Fig. 1 overall quantum circuit schematic for DDQAOA solving an CSPP instance.

Algorithm 1 The algorithm represent the Pseudo code of DDQAOA.

Input: $p_0 = 1$; $p_{\max}$; H_C , $H_M \in \mathbb{R}^n$: Energy improvement threshold $\sigma > 0$; Variance threshold for convergence $k \in \mathbb{N}$; Patience $N_{\text{opt_max}}$; $\text{cost_function}(\gamma, \beta)$

```

1   $p \leftarrow p_0$  Initialize  $\gamma, \beta$  for  $p_0$  Initialize optimizer  $\mathcal{O}$   $E_{\text{hist}} \leftarrow []$ ,
    $E_{\text{best}} \leftarrow \infty$ ,  $\gamma_{\text{best}} \leftarrow \gamma$ ,  $\beta_{\text{best}} \leftarrow \beta$   $c \leftarrow 0$  for  $t = 1$  to  $N_{\text{opt\_max}}$  do
2     $(\gamma, \beta) \leftarrow \mathcal{O}.\text{step}(\text{cost\_function}, \gamma, \beta)$   $E_t \leftarrow$ 
    $\text{cost\_function}(\gamma, \beta)$  Append  $E_t$  to  $E_{\text{hist}}$ 
3    if  $E_t < E_{\text{best}} - \epsilon$  then
4       $E_{\text{best}} \leftarrow E_t$ ,  $\gamma_{\text{best}} \leftarrow \gamma$ ,  $\beta_{\text{best}} \leftarrow \beta$   $c \leftarrow 0$ 
5    else
6       $c \leftarrow c + 1$ 
7    end
8    if  $c \geq k$  or  $\text{Var}(E_{\text{hist}}[-[k/2]:]) < \sigma$  // Plateau
   (Eq. 18) or variance convergence (Eq. 19)
9    then
10     if  $p = p_{\max}$  then
11       break
12      $p_{\text{old}} \leftarrow p$   $p \leftarrow p + 1$   $\gamma_{\text{old}} \leftarrow \gamma_{\text{best}}$ ,  $\beta_{\text{old}} \leftarrow \beta_{\text{best}}$  if
    $p_{\text{old}} = 1$  then
13        $\gamma \leftarrow [\gamma_{\text{old}}, 1.2 * \gamma_{\text{old}}]$   $\beta \leftarrow [\beta_{\text{old}}, 0.8 * \beta_{\text{old}}]$ 
14     else
15        $\text{kind} \leftarrow \text{'cubic'}$  if  $p_{\text{old}} \geq 4$  else  $\text{'linear'}$ 
16        $\gamma \leftarrow \text{interpolate}(\gamma_{\text{old}}, p, \text{kind})$   $\beta \leftarrow$ 
    $\text{interpolate}(\beta_{\text{old}}, p, \text{kind})$ 
17     end
18     Re-initialize optimizer  $\mathcal{O}$   $c \leftarrow 0$ 
19   end
20    $E_{\text{best}} \leftarrow \text{cost\_function}(\gamma, \beta)$   $\gamma_{\text{best}} \leftarrow \gamma$ ,  $\beta_{\text{best}} \leftarrow \beta$ 
21 return  $(\gamma_{\text{best}}, \beta_{\text{best}})$ ,  $E_{\text{best}}$ ,  $p$ ;

```

4 Evaluation

We carefully and rigorously evaluated the performance of DDQAOA against standard fixed-depth QAOA protocols, ensuring a fair and comprehensive comparison.

4.1 Experimental setup

We created a testbed consisting of 100 randomly generated CSPP instances for 10-qubit and 16-qubit problem sizes, and 20 instances on complete 5-node graphs whose QUBO encodings require exactly 22 qubits (instances selected such that the resource limit satisfies $r_{\text{limit}} \in \{2, 3\}$, fixing the slack register at two qubits). The generation process was designed to produce a diverse set of problems to test the algorithms' robustness. All hyperparameters and optimizer settings are identical across the three problem sizes. The generation process was designed to produce a diverse set of problems to test the algorithms' robustness. We then benchmarked DDQAOA against four standard fixed-depth QAOA implementations with circuit depths of $p = 3, 5, 10$, and 15. These baselines represent a range of choices that a practitioner might make, from shallow, resource-efficient circuits to deep, expressive ones.

All experiments were conducted using a PennyLane quantum simulator. We used Adam optimizer for the classical optimization loop. We do not consider the run time or the time-to-solution as performance measures since the timing results obtained from the simulator may not be representative of QAOA performed on a real device.

4.2 Evaluation metrics

We assessed the solution quality using two complementary metrics:

- *Success Probability*: The probability of measuring the true ground state (the optimal solution).
- *Approximation Ratio (r)*: The ratio of the final expectation value to the true minimum eigenvalue of the Hamiltonian, $r = \langle H_{\text{CSPP}}^* \rangle / E_{\text{min}}$, normalized to the range $[0, 1]$. An ideal algorithm achieves $r = 1$.

5 Results

Table 1 provides a summary of our results. In the following, we provide detailed discussions of these results.

5.1 Solution quality

The analysis of solution quality (Approximation Ratio and Success Probability) reveals a clear advantage for

Table 1 Comparative Performance Summary of DDQAOA versus Fixed-Depth QAOA ($p = 3, 5, 10, 15$). The runtime metric is excluded because the simulator timing do not reflect real device performance. We did not perform QAOA optimization on physical hardware. Metrics (success probability) and (approximation ratio) require ground state knowledge and serve primarily for benchmarking; in practical applications, only (expectation value) provides actionable information without prior solution knowledge. The 10- and 16-qubit benchmarks comprise 100 instances each; the 22-qubit benchmark comprises 20 instances, and its values are reported to four decimal places (best value per column in bold)

Method	Qubits	Mean Approx. Ratio (σ)	Median Approx. Ratio	Mean Success Prob. (σ)	Median Success Prob.
DDQAOA	10	0.969 (0.011)	0.973	0.024 (0.025)	0.017
$p = 15$	10	0.953 (0.025)	0.958	0.016 (0.032)	0.004
$p = 10$	10	0.937 (0.028)	0.944	0.011 (0.019)	0.006
$p = 5$	10	0.912 (0.068)	0.923	0.009 (0.009)	0.002
$p = 3$	10	0.869 (0.119)	0.892	0.010 (0.016)	0.006
DDQAOA	16	0.990 (0.003)	0.991	0.006 (0.006)	0.004
$p = 15$	16	0.985 (0.006)	0.986	0.004 (0.005)	0.002
$p = 10$	16	0.981 (0.007)	0.982	0.004 (0.006)	0.002
$p = 5$	16	0.976 (0.010)	0.976	0.003 (0.006)	0.001
$p = 3$	16	0.965 (0.014)	0.966	0.003 (0.006)	0.001
DDQAOA	22	0.9960 (0.0008)	0.9961	0.0124 (0.0047)	0.0130
$p = 15$	22	0.9968 (0.0007)	0.9968	0.0132 (0.0129)	0.0103
$p = 10$	22	0.9963 (0.0008)	0.9963	0.0127 (0.0103)	0.0107
$p = 5$	22	0.9946 (0.0018)	0.9948	0.0114 (0.0073)	0.0098
$p = 3$	22	0.9823 (0.0066)	0.9840	0.0058 (0.0024)	0.0059

DDQAOA over standard QAOA at the 10- and 16-qubit scales, and quality parity with the best fixed-depth circuits at the 22-qubit scale. Figure 2 shows the distribution of approximation ratios across the 100 test instances for both 10-qubit (Fig. 2a) and 16-qubit (Fig. 2b) illustrates convergence throughout the optimization process.

This trend is even more pronounced in the 16-qubit results. Although all methods perform well (ratios ≥ 0.95), DDQAOA again achieves the highest median approximation ratio (≈ 0.991) and the smallest interquartile range,

surpassing even $p = 15$ QAOA. Its standard deviation ($\sigma = 0.003$) is evidently lower than that of any fixed-depth method (σ from 0.006 to 0.014), confirming its ability to reliably find high-quality solutions. Figure 3b shows the convergence. The results showed that a high-depth QAOA circuit achieves a higher approximation ratio than a low-depth QAOA circuit; however, it also requires more computational resources. DDQAOA increases ansatz complexity (i.e., p) only when the optimization landscape requires it; thus, the algorithm benefits from increasing depth while minimizing the computational cost associated with a preset high depth.

The success probability metric, shown in Fig. 4, demonstrates a similar performance. For both 10-qubit (Fig. 4a) and 16-qubit (Fig. 4b) problems, DDQAOA achieves a substantially higher median success probability (for 10-qubit 0.017 and 0.004 for 16-qubit) than any fixed-depth protocol (for 10-qubit from 0.004 to 0.006 and for 16-qubit from 0.001 to 0.002). This indicates a greater capacity to identify the exact ground state, a critical advantage for applications requiring optimality. DDQAOA consistently produces fewer extremely low-value results, demonstrating superior median performance, consistency, and peak success rates. The convergence is shown in Figs. 3c and 3d.

We note that the absolute success probabilities in Table 1 are small because this metric measures the probability of sampling the exact ground state among the 2^N computational basis states ($N = 10$ or 16), a deliberately stringent criterion, and not a deficiency of the classical optimizer. Two factors explain the values. First, the QAOA cost function optimizes the expectation value (energy) rather than the ground-state overlap directly, so a state with a high approximation ratio ($r \approx 0.97$ – 0.99) can still distribute amplitude across many low-energy bitstrings. Second, the Hilbert space grows exponentially with N , so the same degree of amplitude concentration yields a smaller per-state probability at 16 qubits than at 10. Crucially, the measured success probabilities remain well above the uniform random-sampling baseline of $1/2^N$: for DDQAOA the median success probability is 0.017 at 10 qubits ($\approx 17\times$ the $1/1024$ random baseline) and 0.004 at 16 qubits ($\approx 260\times$ the $1/65,536$ random baseline). Because this behavior is intrinsic to the metric and the exponential state space, it affects all methods equally, and DDQAOA retains a consistent success-probability advantage over every fixed-depth baseline (Fig. 4). To examine how the comparison evolves with system size, we further evaluate all methods on the 20 CSPP instances at 22 qubits (Figs. 2c and 4c, Table 1). Three observations

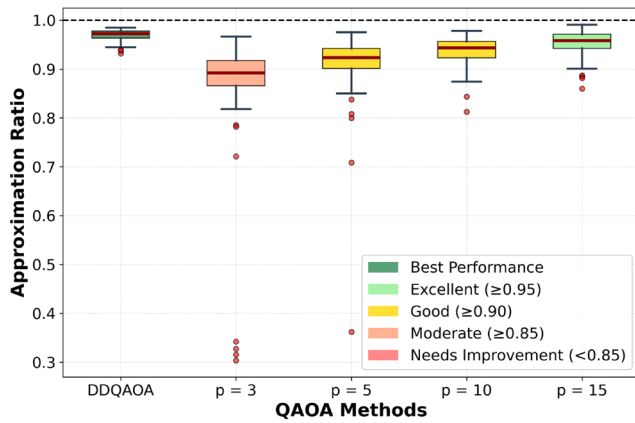
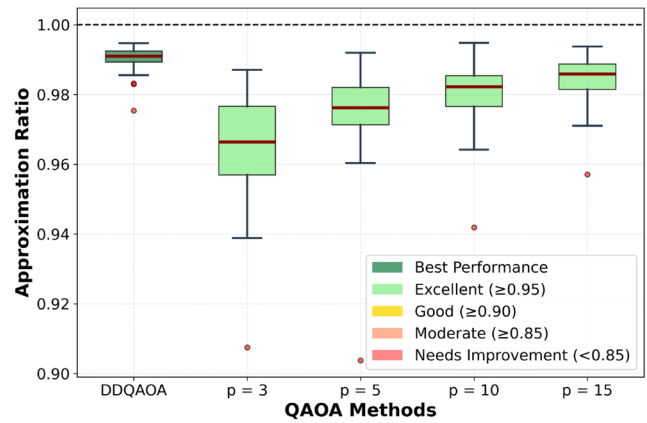
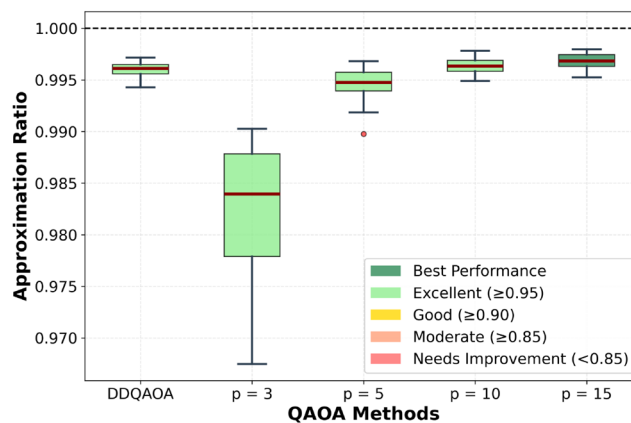
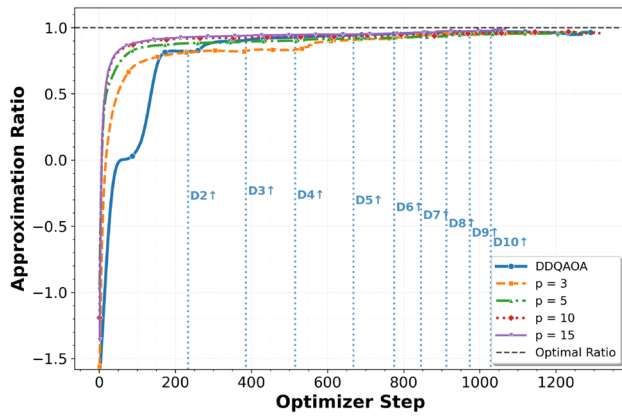
(a) $N=10$ (b) $N=16$ (c) $N=22$

Fig. 2 The box plot shows the approximation ratio distributions across 100 problem instances achieved by DDQAOA and fixed-depth QAOAs. Black dashed lines indicate the optimal approximation ratio line. Panels (a) and (b) show 100 instances at 10 and 16 qubits; panel

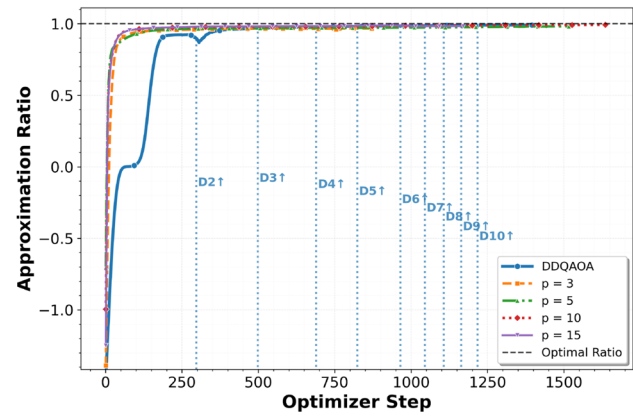
(c) shows 20 instances at 22 qubits. DDQAOA substantially exceeds all fixed-depth baselines at 10 and 16 qubits; at 22 qubits it remains statistically indistinguishable from $p = 10$ and within 0.1% of $p = 15$

summarize the outcome. First, DDQAOA's solution quality does not degrade with size: its mean approximation ratio rises from 0.969 (10 qubits) to 0.990 (16 qubits) to 0.9960 (22 qubits), and it continues to outperform the shallow baselines decisively, exceeding $p = 3$ and $p = 5$ on 20 out of 20 instances (paired Wilcoxon signed-rank test, $p \approx 2 \times 10^{-6}$). Second, at 22 qubits the deep fixed-depth circuits close the gap: DDQAOA is statistically indistinguishable from $p = 10$ (Wilcoxon $p = 0.22$; DDQAOA higher on 10 of 20 instances) and lies within 0.1% of $p = 15$, which attains a marginally higher mean ratio (0.9968 vs. 0.9960, $p \approx 10^{-5}$). We attribute the narrowing of the raw ratio gap partly to a ceiling effect—at 22 qubits every method exceeds $r = 0.98$,

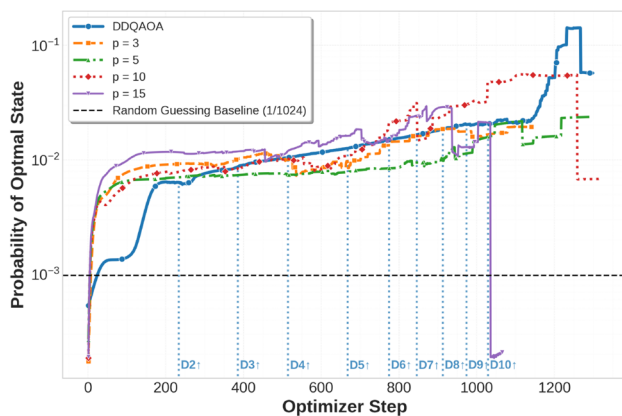
compressing absolute differences near the optimum—and partly to the growing expressivity benefit of deep circuits at larger problem sizes. Third, DDQAOA attains the highest median success probability of all methods at 22 qubits (0.0130, vs. 0.0103–0.0107 for $p = 10$ and $p = 15$), extending the success-probability advantage observed at 10 and 16 qubits, although the fixed-depth distributions exhibit heavier upper tails. Taken together, the advantage of DDQAOA at larger scales is best characterized not as superiority over the best-tuned deep circuit, but as matched solution quality achieved without any a priori depth selection; the resource implications are quantified in the following subsection.



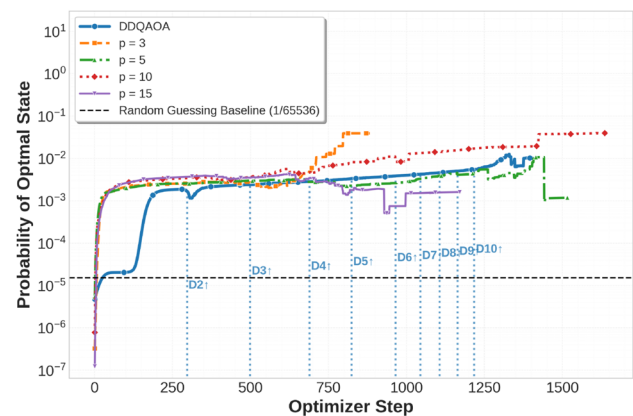
(a) 10-qubit approximation ratio



(b) 16-qubit approximation ratio



(c) 10-qubit success probability



(d) 16-qubit success probability

Fig. 3 It shows the convergence trajectories for (a), (b) approximation ratio, and (c), (d) success probability for 10 and 16 qubits. Vertical dashed lines indicate the increase in layer number in DDQAOA

throughout the optimization, and the horizontal black dashed line indicates the optimal value for the approximation ratio and the baseline for random guessing success probability.

5.2 Resource cost

The comparison of DDQAOA with fixed-depth QAOA must take both solution quality and circuit-resource cost into account. Circuit depth and cumulative two-qubit gate count have been established as primary metrics for assessing the practicality of variational algorithms on NISQ hardware (Zhang et al. 2025). We therefore report cumulative CNOT usage alongside approximation ratio in Table 2, using DDQAOA as the reference baseline. The analysis shows that DDQAOA attains the highest mean approximation ratio among all fixed-depth baselines at 10 qubits (0.969) and 16 qubits (0.990) while using 2.11–3.17 $\times$ fewer cumulative CNOT evaluations than $p = 10$ and $p = 15$ QAOA at 10 qubits, and 1.73–2.59 $\times$ fewer at 16 qubits.

Figure 5 demonstrates the CNOT gate evolution across optimization steps for both DDQAOA and fixed-depth QAOA implementations. For the 10-qubit instance, DDQAOA increases CNOT gate counts from 90 to 900, and the fixed-depth QAOA uses 270, 450, 900, and 1,350 CNOT gates for $p = 3, 5, 10$, and 15, respectively. Similarly, the 16-qubit case shows DDQAOA CNOT gates scaling from 240 to 2,400, whereas fixed-depth QAOA uses 720, 1,200, 2,400, and 3,600 CNOT gates per circuit for $p = 3, 5, 10$, and 15, respectively (a fully connected 16-qubit cost layer contains 120 two-qubit ZZ terms, i.e., 240 CNOT gates). For the 22-qubit instances, each cost layer contains 227–231 ZZ terms (454–462 CNOT gates); DDQAOA grows from one layer (≈ 460 CNOTs) to ten layers ($\approx 4,600$ CNOTs), whereas fixed-depth QAOA uses approximately 1,386,

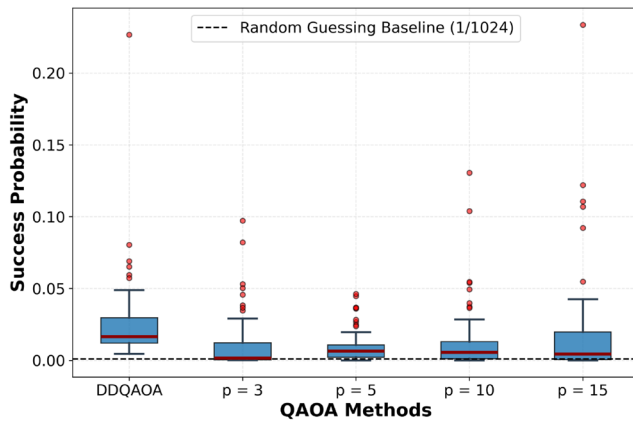
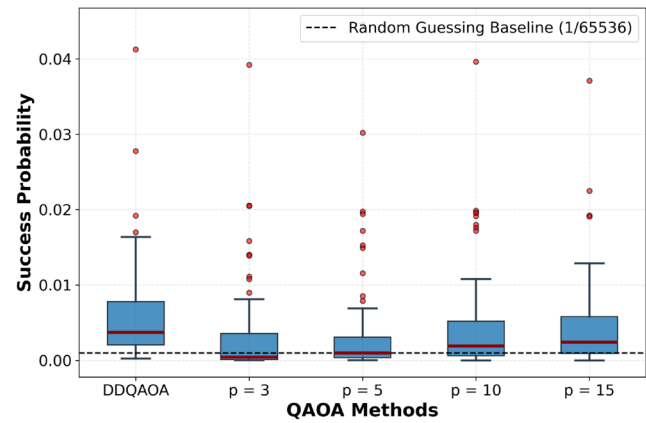
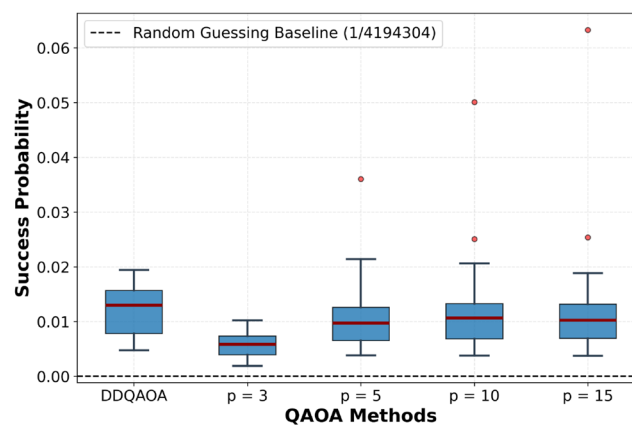
(a) $N = 10$ qubits(b) $N = 16$ qubits(c) $N = 22$ qubits

Fig. 4 The box plot shows the success probability distributions across 100 problem instances for DDQAOA and fixed-depth QAOAs. Black dashed lines indicate a random guessing baseline. Panels (a) and (b) show 100 instances at 10 and 16 qubits; panel (c) shows 20 instances

at 22 qubits. DDQAOA substantially exceeds random performance at every size and attains the highest median success probability among all methods at 22 qubits

2,310, 4,620, and 6,930 CNOT gates per circuit for $p = 3$, 5, 10, and 15 (Fig. 5c).

DDQAOA begins with minimal circuit resources and incrementally adds QAOA layers throughout the optimization process, thereby increasing the CNOT gate count as needed. This makes DDQAOA more suited for NISQ devices, where shallow circuits minimize error accumulation from gate errors and qubit decoherence. The stepwise expansion pattern shows that the number of CNOT gates scales proportionally to the need, thereby avoiding resource waste by allocating only the necessary circuit depth. In contrast, fixed-depth QAOA variants maintain a constant gate count regardless of solution quality or convergence status. Conversely, DDQAOA prioritizes solution accuracy by expanding QAOA circuit layers on demand, based on convergence behavior.

From a cumulative resource perspective, the 10-qubit DDQAOA consumes 511020 CNOT gates (gate count multiplied by optimization steps). Compared to this baseline, $p = 3$ requires 36.6% fewer cumulative gates (324,000), $p = 5$ uses 5.7% more gates (540,000), $p = 10$ uses 111.3% more gates (1,080,000), and $p = 15$ uses 217% more gates (1,620,000). For the 16-qubit case, DDQAOA uses 2,082,720 CNOT gates, whereas $p = 3$ uses 48.1% fewer gates (1,080,000), $p = 5$ requires 13.6% fewer gates (1,800,000), $p = 10$ uses 72.9% more gates (3,600,000), and $p = 15$ uses 159.3% more gates (5,400,000). For the 22-qubit case, computed under the same matched-window accounting, DDQAOA consumes 2,840,838 cumulative CNOT gates, whereas $p = 3$ uses 17.1% fewer gates (2,356,200) at significantly lower solution quality, $p = 5$ uses 38.2%

Table 2 Cumulative CNOT gate usage and solution quality for DDQAOA and fixed-depth QAOA baselines on 10-, 16-, and 22-qubit CSPP instances. Cumulative CNOTs are computed as per-step gate count multiplied by total optimizer iterations. The Overhead is reported relative to the DDQAOA baseline

Method	Qubits	Cumulative CNOTs	r	Overhead vs DDQAOA
DDQAOA	10	511,020	0.969	1.00× (baseline)
$p = 3$	10	324,000	0.869	0.63× (lower r)
$p = 5$	10	540,000	0.912	1.06×
$p = 10$	10	1,080,000	0.937	2.11×
$p = 15$	10	1,620,000	0.953	3.17×
DDQAOA	16	2,082,720	0.990	1.00× (baseline)
$p = 3$	16	1,080,000	0.965	0.52× (lower r)
$p = 5$	16	1,800,000	0.976	0.86× (lower r)
$p = 10$	16	3,600,000	0.981	1.73×
$p = 15$	16	5,400,000	0.985	2.59×
DDQAOA	22	2,840,838	0.996	1.00× (baseline)
$p = 3$	22	2,356,200	0.982	0.83× (lower r)
$p = 5$	22	3,927,000	0.995	1.38×
$p = 10$	22	7,854,000	0.996	2.76×
$p = 15$	22	11,781,000	0.997	4.15×

more (3,927,000), $p = 10$ uses 176% more (7,854,000), and $p = 15$ uses 315% more (11,781,000). The gate advantage of DDQAOA over the deepest baseline therefore persists across all three system sizes (3.17×, 2.59×, and 4.15× at 10, 16, and 22 qubits).

5.3 Parameter evolution

Figures 6a and 6b show the final optimal γ^* of cost and β^* of mixer Hamiltonian for both 10-qubit and 16-qubit instances, respectively. For fixed-depth QAOA ($p = 3, 5, 10, 15$), the optimized parameters exhibit a seemingly random pattern across the parameter indices. In contrast, DDQAOA exhibits monotonic trends across parameter indices that align closely with predictions from adiabatic theory. We emphasize that the parameter patterns shown in Fig. 6 represent locally optimized values at the final converged depth, seeded by the transfer strategy, rather than proven global optima. However, the fact that these patterns emerge consistently across 100 diverse problem instances – with the classical optimizer unconstrained after initialization – provides empirical evidence that the interpolation strategy guides the optimization toward adiabatically structured solution manifolds.

The γ^* parameters show a monotonically increasing pattern across parameter indices, with values consistently

growing from near-zero at initial layers to higher magnitudes at later layers. This parameter value behaviour aligns with the adiabatic evolution, where the cost Hamiltonian coefficient should increase toward the last layer ($p - 1$) to prioritize exploitation of the cost Hamiltonian. Conversely, β^* shows a convergence to zero trend, decreasing from higher initial values to approximately zero at the final layer. This pattern also aligns with the adiabatic conditions that promote exploration in the initial layers, while minimizing mixing in later layers to preserve the evolved state in low-energy configurations near the cost Hamiltonian eigenspace (Wurtz and Love 2022). These structured parameter patterns are consistently observed across both 10-qubit and 16-qubit instances, demonstrating the robustness and scalability of DDQAOA (Ruan et al. 2025). The adherence of DDQAOA parameters to adiabatic principles suggests that the algorithm effectively approximates the optimal adiabatic passage, achieving parameter configurations that would typically require extensive optimization in standard QAOA formulations.

6 Conclusion

QAOA has been extensively utilized to solve various problems, and numerous variants of QAOA have been proposed to enhance its performance. In this paper, we introduce DDQAOA, an optimization QAOA variant that iteratively increases the QAOA depth based on the cost convergence. We tested the effectiveness of DDQAOA on the CSPP, an NP-hard problem, and tested several instances of it. To examine the performance of DDQAOA, we compared the results against the fixed-depth QAOA with $p = 3, 5, 10$, and 15.

The experimental results show that DDQAOA outperforms all fixed-depth baselines at the 10- and 16-qubit scales, and matches the solution quality of the best fixed depths at the 22-qubit scale (statistically indistinguishable from $p = 10$ and within 0.1% of $p = 15$) while attaining the highest median success probability and a persistent cumulative-CNOT advantage over the deepest baseline under matched optimization windows. The method's key advantage lies in its ability to autonomously assign QAOA-layer when needed, avoiding both the underparametrization of shallow QAOA circuits and the wasteful overhead of unnecessarily deep ones. The consistently low standard deviation in approximation ratios across diverse problem instances demonstrates that DDQAOA is a reliable algorithm, crucial for practical applications.

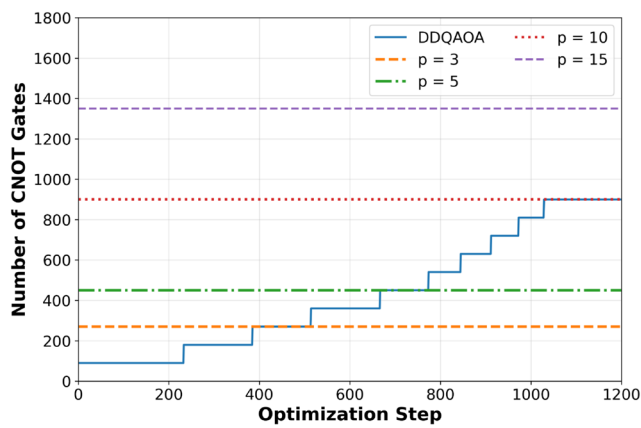
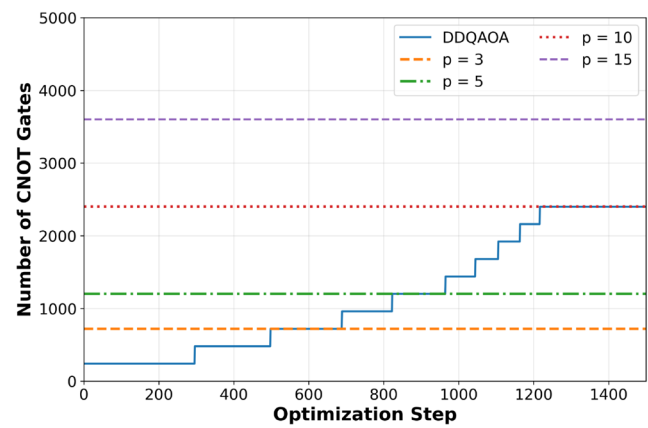
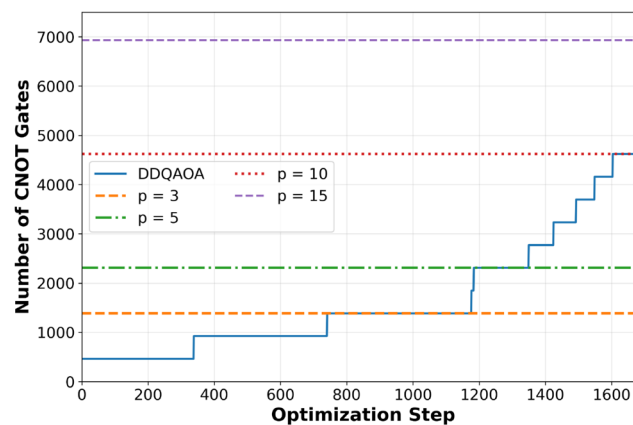
(a) $N = 10$ qubits(b) $N = 16$ qubits(c) $N = 22$ qubits

Fig. 5 It illustrates the number of CNOT gates in the QAOA circuit at each optimization step. The fixed-depth QAOAs show a constant CNOT utilization throughout the optimization; the DDQAOA shows

an increase in the number of CNOT gates, starting from a minimum and reaching equal to the number of CNOT gates utilized by $p = 10$ QAOA. Panel (c) shows a representative 22-qubit instance

A particularly compelling finding is the monotonic increase in γ^* and the decrease in β^* learned by DDQAOA, which can be interpreted as a discretized version of adiabatic evolution. This provides powerful physical intuition for the algorithm's effectiveness: rather than searching randomly through parameter space, DDQAOA discovers a smooth, efficient "annealing schedule" that guides the quantum state from the initial uniform superposition towards the ground state encoding the optimal solution.

Furthermore, DDQAOA enhances optimization efficiency, converging to high-quality solutions with fewer QAOA layers, resulting in a lower total CNOT gate count. This reduction in quantum resources makes the algorithm

more NISQ-aware. By eliminating QAOA depth tuning while improving performance, DDQAOA represents a step toward making QAOA a more robust and practical tool for real-world combinatorial optimization.

While these results are promising, the experiments were conducted on classical simulators for systems up to 22 qubits. Future work will focus on solving larger problem instances and validating the performance on real quantum hardware to assess noise resilience and connectivity constraints. Additional algorithmic refinements could also be explored, such as alternative convergence criteria, more sophisticated parameter transfer techniques, and generalization of DDQAOA to other combinatorial optimization problems.

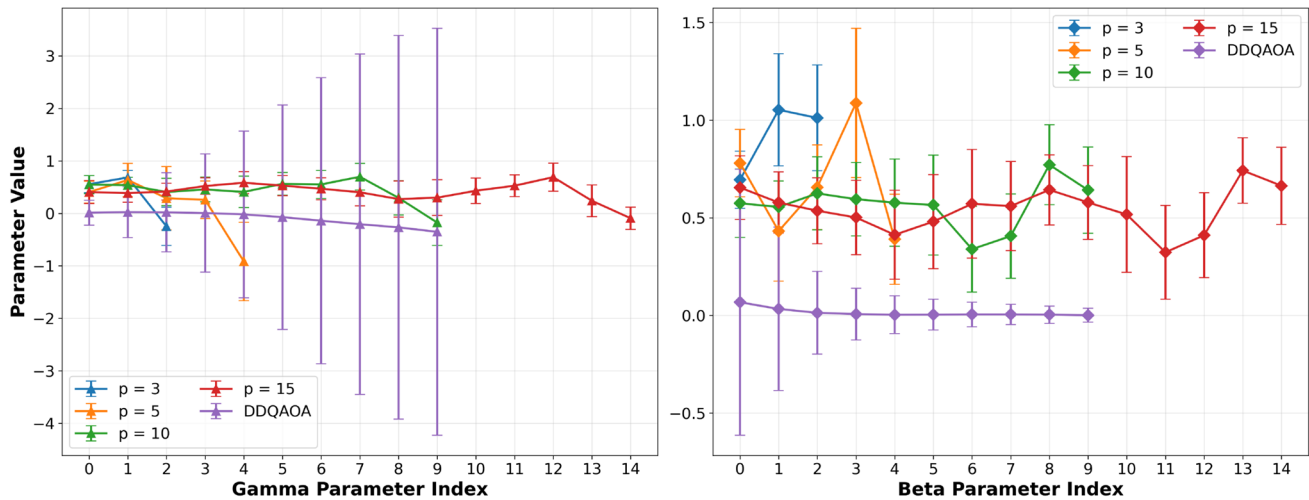
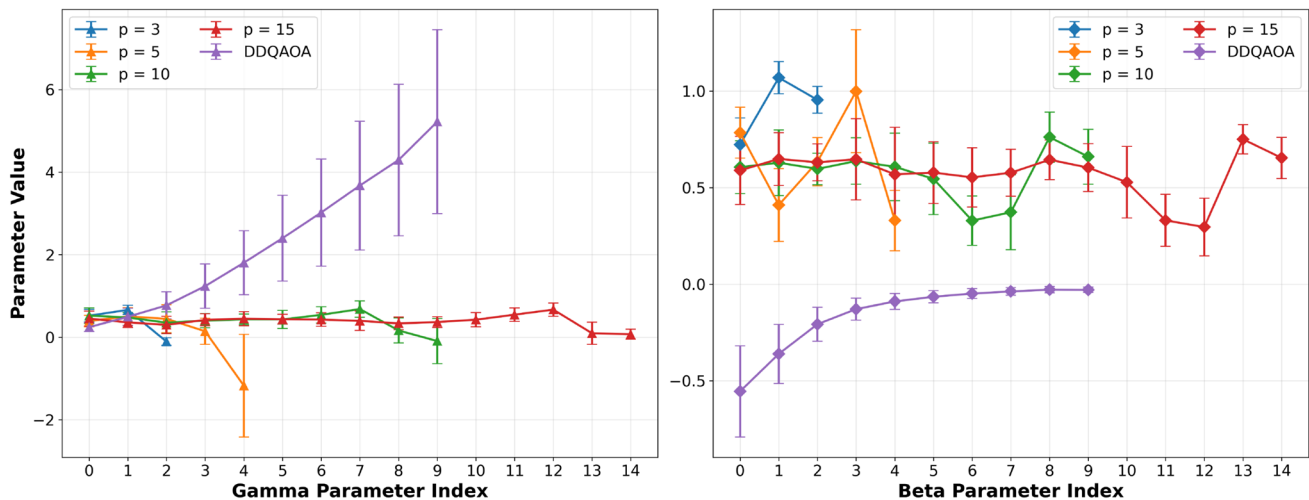
(a) $N = 10$ qubits(b) $N = 16$ qubits

Fig. 6 Optimal parameter profiles for DDQAOA and standard QAOA at their final converged depths. Left panels show γ parameters (cost Hamiltonian); right panels show β parameters (mixer Hamiltonian). Error bars represent standard deviation across 100 problem instances. The DDQAOA parameters shown are the result of full local optimization at the final converged depth, initialized via the interpolation-based transfer strategy (Eqs. 20–22 and Section 3). The monotonic trends

are not imposed by construction – the classical optimizer is free to deviate from the interpolated initialization at each depth – but emerge consistently across instances, suggesting that the transfer scheme seeds the optimization in basins of attraction that correspond to adiabatically structured schedules. The parameters are locally optimized values and are not proven global optima

Author Contributions R.S. developed the methodology, performed experiments, and wrote the original manuscript draft. N.M., S.K. and A.F. reviewed and edited the manuscript. S.K. and A.F. supervised the project, provided critical feedbacks, and verified the results. All authors reviewed and approved the final manuscript.

Funding Open Access funding provided by the Qatar National Library. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data Availability No datasets were generated or analysed during the current study.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Akshay V, Philathong H, Campos E, Rabinovich D, Zacharov I, Zhang X-M, Biamonte JD (2022) Circuit depth scaling for quantum approximate optimization. *Phys Rev A* 106(4):042438
- Arora S, Barak B (2009) *Computational Complexity: a Modern Approach*. Cambridge University Press, Cambridge
- Arute F, Arya K, Babbush R, Bacon D, Bardin JC, Barends R, Biswas R, Boixo S, Brandao FG, Buell DA et al (2019) Quantum supremacy using a programmable superconducting processor. *Nature* 574(7779):505–510
- Bertsekas DP, Tsitsiklis JN (1991) An analysis of stochastic shortest path problems. *Math Oper Res* 16(3):580–595
- Bharti K, Cervera-Lierta A, Kyaw TH, Haug T, Alperin-Lea S, Anand A, Degroote M, Heimonen H, Kottmann JS, Menke T et al (2022) Noisy intermediate-scale quantum algorithms. *Rev Mod Phys* 94(1):015004
- Blekos K, Brand D, Ceschini A, Chou C-H, Li R-H, Pandya K, Summer A (2024) A review on quantum approximate optimization algorithm and its variants. *Phys Rep* 1068:1–66
- Cerezo M, Arrasmith A, Babbush R, Benjamin SC, Endo S, Fujii K, McClean JR, Mitarai K, Yuan X, Cincio L et al (2021) Variational quantum algorithms. *Nat Rev Phys* 3(9):625–644
- Chancellor N, Zohren S, Warburton PA (2017) Circuit design for multi-body interactions in superconducting quantum annealing systems with applications to a scalable architecture. *NPJ Quantum Inf* 3(1):21
- Cheng L, Chen Y-Q, Zhang S-X, Zhang S (2024) Quantum approximate optimization via learning-based adaptive optimization. *Commun Phys* 7(1):83
- Crooks GE (2018) Performance of the quantum approximate optimization algorithm on the maximum cut problem. [arXiv:1811.08419](https://arxiv.org/abs/1811.08419). arXiv preprint
- Daley AJ, Bloch I, Kokail C, Flannigan S, Pearson N, Troyer M, Zoller P (2022) Practical quantum advantage in quantum simulation. *Nature* 607(7920):667–676
- Derigs U, Friederichs S, Schäfer S (2009) A new approach for air cargo network planning. *Transp Sci* 43(3):370–380
- Ebadi S, Keesling A, Cain M, Wang TT, Levine H, Bluvstein D, Semeghini G, Omran A, Liu J-G, Samajdar R et al (2022) Quantum optimization of maximum independent set using rydberg atom arrays. *Science* 376(6598):1209–1215
- Egger DJ, Mareček J, Woerner S (2021) Warm-starting quantum optimization. *Quantum* 5:479
- Farhi E, Goldstone J, Gutmann S (2014) A quantum approximate optimization algorithm. [arXiv:1411.4028](https://arxiv.org/abs/1411.4028) arXiv preprint
- Gamache M, Soumis F, Marquis G, Desrosiers J (1999) A column generation approach for large-scale aircrew rostering problems. *Oper Res* 47(2):247–263
- Garey MR, Johnson DS (2002) *Computers and Intractability*, vol 29. W. H. freeman, New York
- Goto H, Tatsumura K, Dixon AR (2019) Combinatorial optimization by simulating adiabatic bifurcations in nonlinear hamiltonian systems. *Sci Adv* 5(4):2372
- Graves GW, McBride RD, Gershkoff I, Anderson D, Mahidhara D (1993) Flight crew scheduling. *Manage Sci* 39(6):736–745
- Grimsley HR, Economou SE, Barnes E, Mayhall NJ (2019) An adaptive variational algorithm for exact molecular simulations on a quantum computer. *Nat Commun* 10(1):3007
- Grönkvist M (2006) Accelerating column generation for aircraft scheduling using constraint propagation. *Comput Oper Res* 33(10):2918–2934
- Handler GY, Zang I (1980) A dual algorithm for the constrained shortest path problem. *Networks* 10(4):293–309
- Harrigan MP, Sung KJ, Neeley M, Satzinger KJ, Arute F, Arya K, Atalaya J, Bardin JC, Barends R, Boixo S et al (2021) Quantum approximate optimization of non-planar graph problems on a planar superconducting processor. *Nat Phys* 17(3):332–336
- Korte B, Vygen J (2008) *Combinatorial Optimization: Theory and Algorithms*. Springer, Berlin, Heidelberg
- Lavoie S, Minoux M, Odier E (1988) A new approach for crew pairing problems by column generation with an application to air transportation. *Eur J Oper Res* 35(1):45–58
- Lee X, Yan X, Xie N, Cai D, Saito Y, Asai N (2024) Iterative layerwise training for the quantum approximate optimization algorithm. *Phys Rev A* 109(5):052406
- Long CK, Dalton K, Barnes CH, Arvidsson-Shukur DR, Mertig N (2024) Layering and subpool exploration for adaptive variational quantum eigensolvers: reducing circuit depth, runtime, and susceptibility to noise. *Phys Rev A* 109(4):042413
- Montañez-Barrera J, Michielsen K (2025) Toward a linear-ramp qaoa protocol: evidence of a scaling advantage in solving some combinatorial optimization problems. *NPJ Quantum Inf* 11(1):131
- Muter İ, Birbil Şİ, Bülbül K, Şahin G, Yenigün H, Taş D, Tüzün D (2013) Solving a robust airline crew pairing problem with column generation. *Comput Oper Res* 40(3):815–830
- Pan Y, Tong Y, Yang Y (2022) Automatic depth optimization for a quantum approximate optimization algorithm. *Phys Rev A* 105(3):032433
- Papadimitriou CH, Steiglitz K (1998) *Combinatorial Optimization: Algorithms and Complexity*. Courier Corporation, Mineola, NY
- Preskill J (2018) Quantum computing in the nisq era and beyond. *Quantum* 2:79
- Ramos-Figueroa O, Quiroz-Castellanos M, Mezura-Montes E, Schütze O (2020) Metaheuristics to solve grouping problems: a review and a case study. *Swarm Evol Comput* 53:100643
- Reinhardt LB, Pisinger D (2011) Multi-objective and multi-constrained non-additive shortest path problems. *Comput Oper Res* 38(3):605–616
- Ren Z, Rubinstein ZB, Smith SF, Rathinam S, Choset H (2023) Erca*: a new approach for the resource constrained shortest path problem. *IEEE Trans Intell Transp Syst* 24(12):14994–15005
- Ruan Y, Chen P, Li Q, Yang L, Yuan Z, Xue X, Li X, Liu Z (2025) Toward a linear-ramp QAOA protocol: evidence of a Kibble-Zurek scaling in variational quantum algorithms. *NPJ Quantum Inf* 11(1):41. <https://doi.org/10.1038/s41534-025-01082-1>
- Smith OJ, Boland N, Waterer H (2012) Solving shortest path problems with a weight constraint and replenishment arcs. *Comput Oper Res* 39(5):964–984
- Stojković M, Soumis F, Desrosiers J (1998) The operational airline crew scheduling problem. *Transp Sci* 32(3):232–245
- Sureshbabu SH, Herman D, Shaydulin R, Basso J, Chakrabarti S, Sun Y, Pistoia M (2024) Parameter setting in quantum approximate optimization of weighted problems. *Quantum* 8:1231
- Tang HL, Shkolnikov V, Barron GS, Grimsley HR, Mayhall NJ, Barnes E, Economou SE (2021) qubit-adapt-vqe: An adaptive algorithm for constructing hardware-efficient ansätze on a quantum processor. *PRX Quantum* 2(2):020310
- Tong G (2021) Usco-solver: solving undetermined stochastic combinatorial optimization problems. *Adv Neural Inf Process Syst* 34:1646–1659
- Truger F, Barzen J, Bechtold M, Beisel M, Leymann F, Mandl A, Yusupov V (2024) Warm-starting and quantum computing: a systematic mapping study. *ACM Comput Surv* 56(9):1–31
- Tyagin I, Farag MH, Sherbert K, Shirali K, Alexeev Y, Safo I (2025) Qaoa-gpt: Efficient generation of adaptive and regular quantum approximate optimization algorithm circuits. [arXiv:2504.16350](https://arxiv.org/abs/2504.16350) arXiv preprint

- Vijendran V, Das A, Koh DE, Assad SM, Lam PK (2024) An expressive ansatz for low-depth quantum approximate optimisation. *Quantum Sci Technol* 9(2):025010
- Villeneuve D, Desaulniers G (2005) The shortest path problem with forbidden paths. *Eur J Oper Res* 165(1):97–107
- Wolsey LA, Nemhauser GL (1999) *Integer and Combinatorial Optimization*. John Wiley & Sons, New York
- Wurtz J, Love P (2022) Counterdiabaticity and the quantum approximate optimization algorithm. *Quantum* 6:635. <https://doi.org/10.22331/q-2022-01-27-635>
- Wurtz J, Love P (2021) Maxcut quantum approximate optimization algorithm performance guarantees for $p > 1$. *Phys Rev A* 103(4):042612
- Yanakiev N, Mertig N, Long CK, Arvidsson-Shukur DR (2024) Dynamic adaptive quantum approximate optimization algorithm for shallow, noise-resilient circuits. *Phys Rev A* 109(3):032420
- Yu Y, Cao C, Wang X-B, Shannon N, Joynt R (2023) Solution of sat problems with the adaptive-bias quantum approximate optimization algorithm. *Phys Rev Res* 5(2):023147
- Zhang Z, Paredes R, Sundar B, Quiroga D, Kyriallidis A, Duenas-Osorio L, Pagano G, Hazzard KR (2025) Grover-qaoa for 3-sat: quadratic speedup, fair-sampling, and parameter clustering. *Quantum Sci Technol* 10(1):015022
- Zhong H-S, Wang H, Deng Y-H, Chen M-C, Peng L-C, Luo Y-H, Qin J, Wu D, Ding X, Hu Y et al (2020) Quantum computational advantage using photons. *Science* 370(6523):1460–1463
- Zhou L, Wang S-T, Choi S, Pichler H, Lukin MD (2020) Quantum approximate optimization algorithm: performance, mechanism, and implementation on near-term devices. *Phys Rev X* 10(2):021067
- Zhu L, Tang HL, Barron GS, Calderon-Vargas F, Mayhall NJ, Barnes E, Economou SE (2022) Adaptive quantum approximate optimization algorithm for solving combinatorial problems on a quantum computer. *Phys Rev Res* 4(3):033029
- Zhu Y, Zhang Z, Sundar B, Green AM, Huerta Alderete C, Nguyen NH, Hazzard KR, Linke NM (2023) Multi-round qaoa and advanced mixers on a trapped-ion quantum computer. *Quantum Sci Technol* 8(1):015007
- Zou P (2025) Multiscale quantum approximate optimization algorithm. *Phys Rev A* 111(1):012427

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.